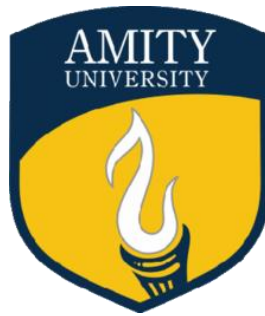


A Project Report
On
**Identification and improvement of process waste in a production line through the
integration of VSM, system dynamics for a complex value stream**

Submitted to
Amity University, Uttar Pradesh



in partial fulfilment of the requirements for the award of the degree of
Bachelor of Technology
in
Mechanical Engineering

by

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NAMAN NARULA

under the guidance of

Dr. Shyamal Samant

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AMITY UNIVERSITY UTTAR PRADESH NOIDA (U.P.)

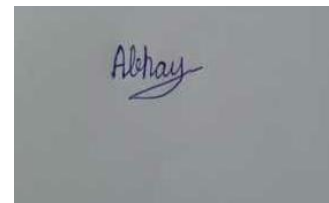
DECLARATION BY THE STUDENT

We, Abhay Tyagi and Naman Narula students of B.Tech (Mechatronics) hereby declare that the project titled “Identification and improvement of process waste in a production line through the integration of VSM and system dynamics for a complex value stream” which is submitted by us to Department of Mechanical Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfilment of requirement for the award of the degree of Bachelor of Technology in Mechanical Engineering, has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

AUUP Noida

Date:

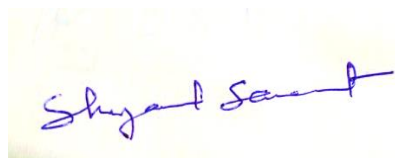
Name and signature of Student(s)

A photograph of a handwritten signature in blue ink on a grey background. The signature is cursive and appears to read 'Abhay'.A photograph of a handwritten signature in blue ink on a white background. The signature is cursive and appears to read 'Naman Narula'.

CERTIFICATE

On the basis of declaration submitted by Abhay Tyagi and Naman Narula, students of B.Tech(Mechatronics), I hereby certify that the project titled “Identification and improvement of process waste in a production line through the integration of VSM and system dynamics for a complex value stream” which is submitted to Department of Mechanical Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida, in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Mechatronics engineering, is an original contribution with existing knowledge and faithful record of work carried out by him/them under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere



AUUP Noida

Date:20-Oct-2023

Dr. Shyamal Samant

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Last but not the least I would like to mention here that I am greatly indebted to all those who has been associated with my project at any stage but whose name does not find place in this acknowledgement.

With regards

Abhay Tyagi

Naman Narula

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ABSTRACT

Value stream mapping is a visual modelling tool aimed at identifying process bottle necks and lean wastes. Vsm carries out a static snapshot of the value stream of a production system, or a supply chain in order to overcome this limitation of vsm system dynamics(sd), modelling approach is used to understand dynamics behaviour of the system, in terms of stocks and flows SD translates the value stream into system dynamics model . an extended VSM methodology and develop a corresponding System Dynamics Model (SDM) for an automotive industry case study. Using Vensim software, they simulate scenarios involving delivering high-quality products, Manpower optimization, , machine failures, and non-conformities. This research demonstrates that the proposed extended VSM can be the sole data source for building a computer model of a manufacturing line. It highlights that integrating VSM and SDA goes beyond just waste identification. It allows for understanding the impact of identified problems on production efficiency, empowering managers to make better decisions.

INTRODUCTION

Companies face challenges in streamlining their value chains to reduce inventory, increase productivity, and shorten delivery times . Lean Manufacturing offers valuable approaches to identify and eliminate waste, thereby improving process efficiency . Value Stream Mapping (VSM) is a widely used Lean technique for this purpose . However, VSM struggles to capture the dynamic nature of production systems .

Integrating computer-based simulation with VSM can provide a more comprehensive understanding of current and future production states . While Discrete Event Simulation (DES) dominates the field , System Dynamics (SD) offers a closed-loop, systemic approach for complex production systems with a supply chain perspective .

Limited research explores the combined use of Lean and SD. Pioneering work by Disney et al. applied SD within Lean Thinking, but without VSM integration. Later studies by Agyapong-Kodua et al. and others used case studies to integrate SD and VSM, aiming to achieve a more dynamic VSM through system thinking and modeling .

Despite these contributions, a standardized framework for translating VSM into quantitative SD models (hard approach) is missing. The existing qualitative (soft) approach in SD utilizes influence diagrams, while the hard approach employs stock and flow diagrams suitable for quantitative simulation. While the symbology between these diagrams and VSM is similar, a standard conversion framework remains elusive.

VSM's popularity stems from its standardized and straightforward modeling process, creating a common language for both practitioners and researchers . To bridge the gap and facilitate SD adoption within the VSM community, a standardized and user-friendly conversion process is necessary.

This research proposes a novel practical framework called VSM-to-SD to assist modellers in efficiently translating VSM into quantitative SD models. The framework leverages the notation of popular modeling tools like Stella® and Vensim® to ensure ease of use. This will empower decision-makers and modellers to develop SD-based VSMs, creating decision support tools for scenario analysis and expanding analytical decision-making capabilities beyond traditional, static VSM. Developing such analytical abilities is crucial for navigating Industry 4.0 and Digital Transformation .

LITERATURE REVIEW

The lean manufacturing originated at Toyota Production System (Ohno, 2019). Value stream mapping emerged as a powerful tool for lean implementation (Rother and Shook, 2003). This section provides a review of lean, VSM. This also details the supporting methodologies that have been used along with VSM.

Antonelli et al. (2018) proposed two kinds of simulation methods: Discrete event simulation (DES) , and System Dynamics for manufacturing system performance evaluation . de Assis et al(2021) provided the framework known as vsm-sd using 2 tools Stella(C) and Vensim(C). Forno et al(2014) used value stream mapping to observe the gaps and consciences of not using it properly. Kibira et al(2009) proposed the sustainable use of system dynamics modelling in social domains, manufacturing, and environmental, financial. Maskell et al(2011) offers a system for determining and also taking care of accountancy techniques effectively concentrating on sensible techniques for monetary administration within companies. Guide intends to assist visitors on carrying out lean accountancy concepts to enhance procedures and also improve general efficiency. Mostafa et al(2013) provided a conceptual framework for lean manufacturing. Ngadono et al(2020) used the value stream mapping for lean manufacturing for one of the biggest local tire manufactures located in Tangerang. Noto and Cosenz(2021) suggested that SDM supports VSM in lean implementation. Rodriguez et al.(2021) modelled a supply chain for additive manufacturing based production line using system dynamics methodology. Samant and Prakash(2021) proposed a framework for lean implementation using using VSM, constraint programming (CPLEX) and simulation for lean implementation. Sundar et al.(2014) presented a map to lean implementation. Verma and Sharma (2016) discussed a method for achieving and lean and green manufacturing by analysis of material and energy flows.. Vahidi proposed a

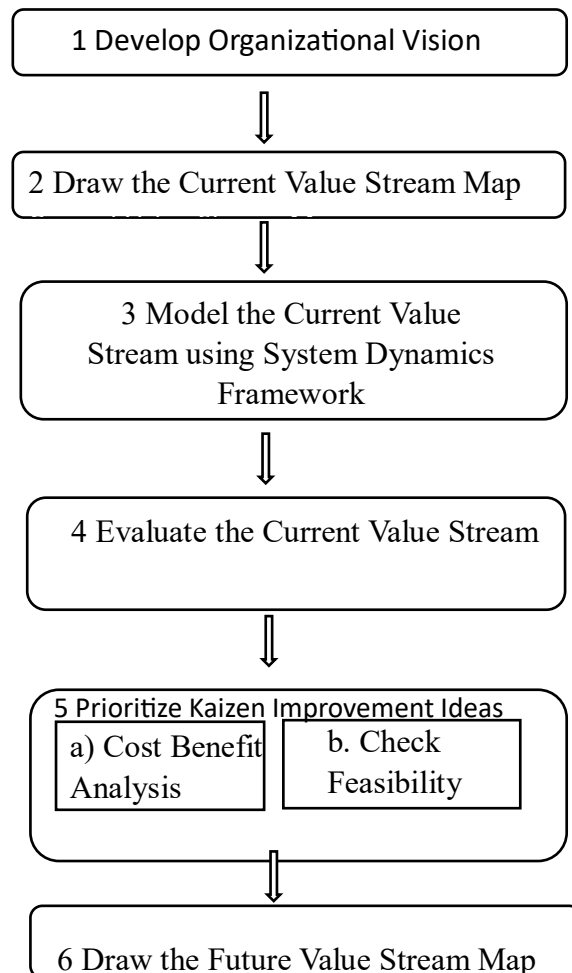
framework using Viable system model for simulating a VSM using system dynamics. Verma and Sharma (2019) presented a literature review of extension of VSM to Energy Value Stream. Zanker et al. (2021) presented a literature review on application of system dynamics to various domains i.e. health, business, and environment. Ohno, T. (2019) gave Toyota Manufacturing System, concentrating on its advancement essential concepts, as well as effect on the sector. It goes over the system's beginnings, the significance of synergy waste removal, plus adjusting to transforming financial problems. The message highlights the change from conventional production techniques to even more reliable as well as considerate techniques. It stresses the requirement for constant enhancement versatility, plus the search of high quality as well as efficiency in the office. Rother et al (2003) shows the Toyota Production System, focusing on its innovation crucial principles, as well as influence on the industry. It reviews the system's origins, the significance of collaboration waste removal plus getting used to transforming financial problems. The message stresses the change from conventional production methods to even more effective and also thoughtful techniques. It highlights the requirement for continual renovation convenience, plus the search of high quality as well as efficiency in the office.

Research Gap:

Value stream mapping is powerful tool for lean implementation such as identifying value-adding activities and waste in a process. However, it is important to be aware of the limitations and challenges associated with it. Limitation of VSM is that it relies heavily on the accuracy and completeness of the data collected. challenge of VSM is the complexity of mapping complex processes. System dynamics employs concepts such as interconnectedness, feedbacks, adaptive capacity/resilience, self-organization, and emergence to help people make better decisions when dealing with complex, dynamic systems .the research proposed a method for combining VSM-based mapping and analysis with SD.

Methodology

The section discusses a framework for lean implementation using value stream map, system dynamics, lean box score etc. The methodology for achieving lean implementation orientation through the proposed approach is depicted in flowchart shown in figure x.1.



Flowchart of the proposed framework

The flowchart (point 1 to 6) is explained using the case of a production line of hot water generator product. The refrigeration equipment manufacturer located in India. The manufacturer produces a range of manufacturing equipment e.g. strainer, butterfly valve, hot water generator, air washers *etc.* The production line involves carrying out the initial assembly, followed by making connections, painting, testing etc. The value stream map for the product is shown in figure y.

1. Develop Organizational Priorities

To achieve lean implementation, it is necessary priorities of the organization are clearly defined. The priorities / goals can be externally focussed, or internally focussed (figure 1). Externally focussed issues are those which involve carrying out changes outside the organization boundary *e.g.* sales, marketing issues. Internally focussed issues are those where the changes must be carried out within the boundary of the organization.

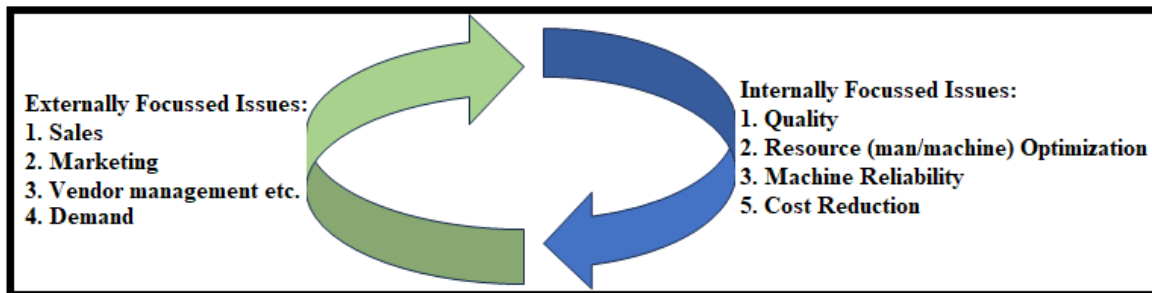


Figure 1: Organizational priorities

This research paper focusses on internally focussed organizational priorities (e.g. quality, resource optimization), as these issues are in direct control of the project team.

2. Draw the Current Value Stream Map (VSM)

Value stream map (VSM) is created for hot water production line (Figure 2). The VSM is a pictorial representation for the series of process steps involved in the production of hot water generator. The benefits of using a value stream map is that it can help to identify areas of waste and inefficiency in the production process. It can help to improve communication and collaboration between different departments. It can help to reduce lead times and improve delivery performance. The creation of value stream map is carried out using “Gemba walk”. During the “Gemba Walk” lean practitioner collects production data at every station *e.g.*, cycle time, machine and manpower resources deployed, defect rate, available time, uptime etc. Inventory data is collected for every stage of the production system. This includes raw material, work-in-progress, finished goods inventory. The lean practitioner takes a pen, pencil and paper and notes down following important parameters at each process step:

Parameters of VSM (within the process icon)

Cycle Time (C/T)

Uptime

Available time

Quality level

Shift time

Changeover Tie (C/O)

(not exhaustive)

Inventory Accumulation: Note down average inventory accumulation between process steps i.e. in between workstations, machining centres, process centres etc. Inventory accumulation expressed in terms of units is converted into number of days of inventory accumulation using the following formula:

Inventory Accumulation (on an average, expressed in days) = $\frac{\text{Inventory Accumulation (on an average, expressed in number of units)}}{\text{(daily demand, expressed in number of units to be produced per day)}}$

Document the process steps: The process steps are documented. The process steps show the raw material entering the first workstation, followed by secondetc. Each of the process steps are documented.

Draw the information flow: Apart from material flows, information flows are also documented. Electronic communication is represented by jogged lines, whereas, manual communication is represented by straight lines.

Creation of a timeline: On the bottom of the VSM, a timeline is added. This includes both values added time (shown at the bottom of the timeline); inventory, which is added at the top of the timeline, is added at the top of the timeline.

A timeline is added at the bottom of the VSM. On this timeline two sets of data are entered. The first set representing the value-added times is entered at the bottom of the timeline.

The process steps as shown in the VSM (Figure 2) related to the hot water generator production line are discussed below (Table 1). The batch size is 4.

Table 1: Process Steps in the value stream map

Step No	Description of the Process Step
1	Customer Places Order (Figure X) Customer order is the starting point of the design activity. The production is carried out using “Made to Order (MTO)” basis.
2	Design Department Creates Product Design (Figure X)

	When the company receives the order from the customer, the design of the hot water is generator is created by the company. The production of the hot water is created using this design.
3	Production Control Activity (Figure X) Production control activity is carried out. Items to be produced in-house: Production order is created for items to be produced in-house. Items to be procured from outside: Procurement of Items which are to be procured from suppliers is carried out.
4	Supplier Supplies the Items (Figure X): Supplier supplies the items to the raw-material warehouse.
5	Angle Cutting (Figure X): This step involves cutting the metal angles that will form the frame of the hot water generator.
6	Arrangement of Base (Figure X): This process involves arranging the base of hot water generator.
7	Outer Shell Layers (Figure X) : This process of assembling the metal shells of the hot water generators.
8	Shell Layering (Figure X) The cut metal angles are arranged and welded together to form the shell of the hot water generator.
9	Initial Assembly (Figure X): In this stage, other key components like the heat exchanger, water tank, and control panel are assembled into the shell.
10	Final Connections (Figure X): Here, any final plumbing or electrical connections are made.
11	Painting (Figure X): The hot water generator is painted to protect it from corrosion and give it a finished look.
12	Testing (Figure X): The completed hot water generator is thoroughly tested to ensure it functions properly and meets safety standards.

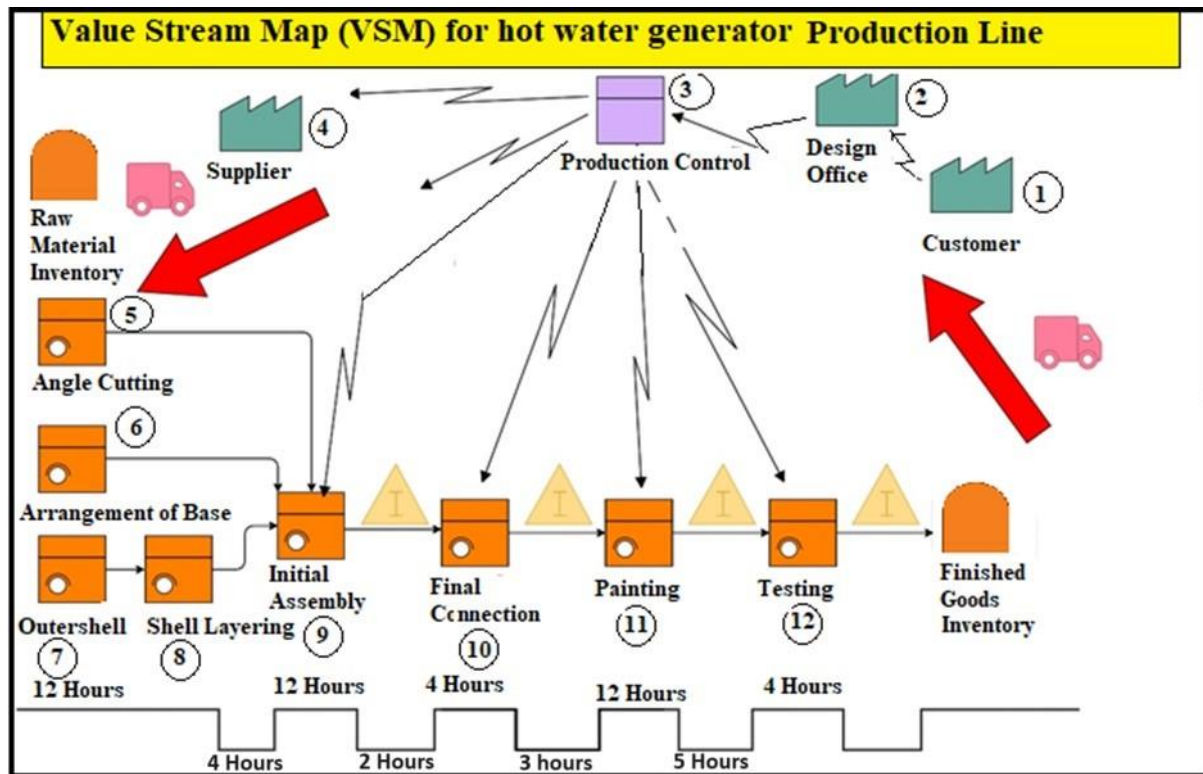


Figure 2: Value Stream Map for the hot water generator production line.

Correct inventory location

The parameters associated with the process steps are given in table 2.

Table 2: Process parameters associated with the VSM

Process Step	CT (Cycle time)	CO (Change over time)	Uptime	Quality
Angle Cutting (5)	6	30 min	35.8	100%
Arrangement of Base (6)	12	30 min	35.8	75%
Outer Shell Layers (7)	4	30 min	35.5	100%
Shell Layering (8)	2	1 hour	35.5	100%
Initial Assembly (9)	12	2-3 hour	35.6	75%
Final Connection (10)	4	30-40 min	-	100%
Painting (11)	12	10 min	35.8	75%
Testing (12)	4	2-3 hr	-	100%

Now, we will discuss the important process parameters associated with the value stream.

Customer demand:

The customer demand drives the production line. Average number of products required per week is four units. Now, takt time must be calculated. The purpose of takt time is match the production rate with external demand.

Takt time:

Takt Time = (Available time for production)/Demand = 36 Hours / 4 units = 9 hours per product.

Available Time per day:

Available time per day (= 7 Hours) - Breaks (1 Hours) = 6 Hours

Cycle time:

Time elapsed between when one unit is produced from a production station to when the next unit is produced. The cycle times at each process step is shown in table X.

Inventory (units):

Inventory accumulation occurs between the two production stations. Inventory accumulation should be minimized, as this reflects blocked capital, space *etc.*

Inventory Accumulation (on an average, expressed in days) = Inventory Accumulation (on an average, expressed in number of units)/(daily demand, expressed in number of units to be produced per day).

Inventory (Number of days) is shown in VSM (Figure 2).

Parallel processes (Figure 3):

When two process steps are parallel, then higher of the two values are taken in the timeline, both in cycle time, and as well as inventory.

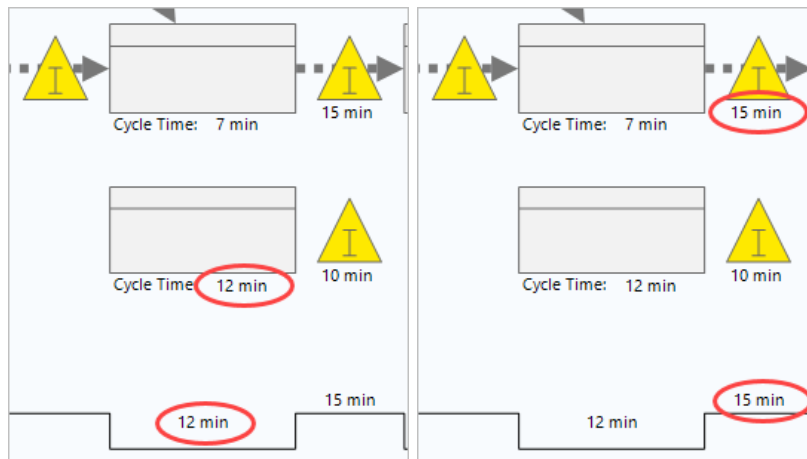


Figure 3: Parallel Process steps (<https://support.minitab.com/en-us/engage/help-and-how-to/tools/value-stream-maps/map-and-timeline-calculations/>)

Lead Time:

This is the time between raw material entering the production line, and exiting the production line, *i.e.* sum of inventory time (between process steps), plus cycle time (process steps) = 58 hours = (12+4+12+2+4+3+12+5+4)hours

Raw Materials Inventory Time

Inventory time for the raw material inventory = 5 days.

Finished Goods Inventory Time

Inventory time for the finished goods inventory (number of days) = 2 days

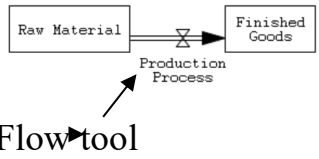
3. Model the current value stream map using systems dynamics framework.

VSM, despite being a very popular tool for lean waste reduction, has several limitations [5][6][7]. VSM provides only a static snapshot of the production system. VSM does not any provide any insight into the production system through dynamic visualization. VSM does include any economic measure of the value stream, E.g., profit, revenue etc. VSM cannot handle complex production value streams. The example of complex value streams includes multiple product scenario, merging flow (E.g., assembly of components), diverging flow (disassembly).

The System Dynamics (SD) approach has emerged as one of the very powerful computer-based tools to build simulated model complex systems to study the behaviour of the systems. The system is modelled through a combination of various sub-systems and components inter-connected through a set of flows,

feedbacks, and variables. The primary building blocks of the system dynamics models are stocks, flows, feedbacks etc. Table 1 gives the primary building blocks used in a systems dynamics model in Vensim software.

Icons (tools) USED IN VENSIM

Icons (Tools in Vensim Software)	Description
 Variable	Defines any variable in the system. It can represent any input, output, intermediate variable etc.
 Stocks	Defines accumulations, or levels in the system. In a production system the stocks can be used to represent inventory.
 Flow tool	Flow tool can be used to model flows in the system. In a production system the flows can be used to represent production process between two inventory locations (stocks).
 Arrow tool	Arrow tool can be used to represent a link between two variables, or between variables and stocks etc.

Sample simulated system dynamics model of a

Sample simulated system dynamics model of a production system is shown in Fig. 4. It shows how various stocks and flows (processes) are related to one another.

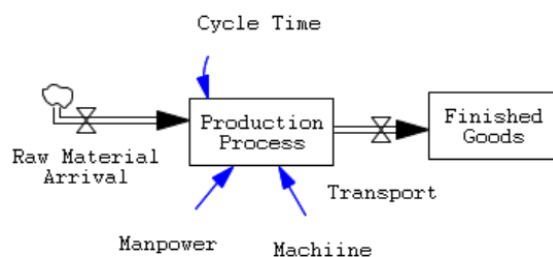


Fig. 4: Sample System Dynamics (Vensim model)

Vensim model for production system is shown in figure 5.

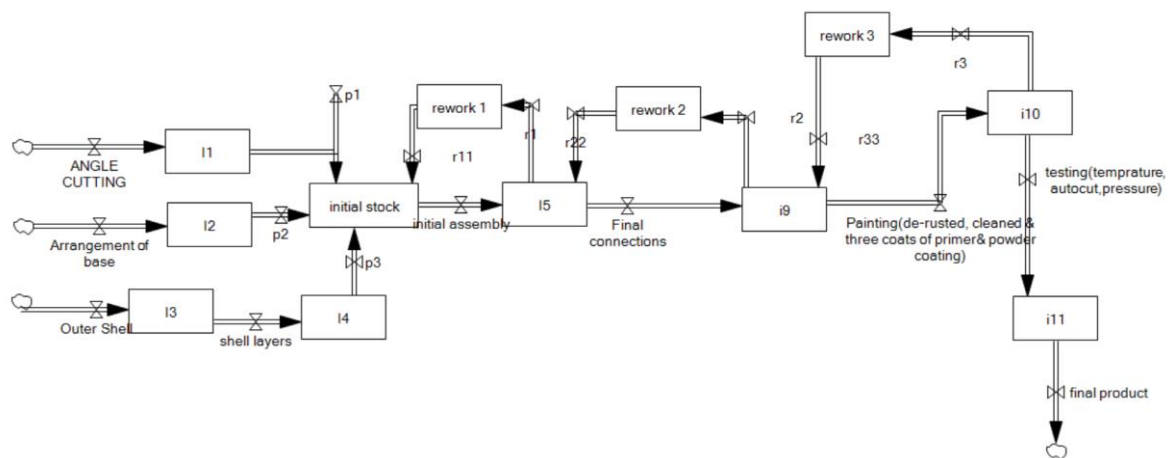


Figure 5: Vensim model of the production system

4. Evaluation of current value stream mapping

The current value stream map is evaluated using lean box score [Maskell et. al, 2011]. The method of lean box score is a comprehensive method of evaluating financial, operational, resource capacity health of the value stream. The lean box score is discussed in table 3.

Table 3: Lean Box Score for the hot water generator value stream map
(Refer to annexure A for the explanation / calculation of the lean box
score(Maskell, 2011))

	Lean Box Score	Box Score Description	Lean Box Score Value
Operational	Dock to Dock days	The amount of time spent from the receipt of material in the raw material warehouse to shipment of the material to the customer.	58hours= (12+4+12+2+4+3+12+5+4)
	First Time Through (FTT)	This parameter measures quality level in the value stream. This denotes the percentage of total products which pass through the production line without being repaired, reworked, scrapped etc.	75%
	On-Time Delivery (OTD)	Percentage of products shipped at the right time	100%
	Floor space	This metric tracks floor space usage	46796 sqft
	Sales per person	Total Sales/Total number of operators employed	Rs 130,000
	Avg cost per unit	Total cost/number of unit produced	55034.46

Resource Capacity	Productive	The shows resource capacity (both manpower, machine) used for actual operations (%). This is expressed as a percentage of the total capacity of that resource.	Paint Guns-0.13
			Drilling nuts-0.66
			Lifting4 machine-0.33
			Welding machine-0.83
			Shearing machine-0.55
			Bending machine-0.41
	Non-productive	The resource capacity (both manpower, machine) used for non-productive operations <i>e.g.</i> inspection, set-up <i>etc.</i> (%).	Paint Guns-0.004
			Drilling nuts-0.002
			Lifting machine-0.016
			Welding machine-0.004
			Shearing machine-0.016
			Bending machine-0.016
	Available	100% – (Productive - Non-productive (%))	Paint Guns-217.5
			Drilling nuts-170
			Lifting machine-41.87
			Welding machine-42.5
			Shearing machine-28.12
			Bending machine-36.87
Financial	Inventory value	Cost of inventory that belongs to the value stream	87.5
	Revenue	Revenue from the shipped products	2,340,000
	Material cost	Material costs	1,100,000
	Conversion cost	Costs incurred in converting the raw materials into final products	Paint Guns-14694.3
			Drilling nuts-4578.84
			Lifting machine-28125.7
			Welding machine-19373
Value stream profit	Revenue – Costs	Shearing machine-10554.6	
		Bending machine-10512.4	
FOR CALCULATION REFER TO Appendix A			

5. Kaizen improvement

Kaizen improvement is a process of systematic improvement, where the lean practitioner studies the value stream in detail, and comes up with continuous improvement. These improvement efforts can come up in a variety of areas e.g. quality, reliability, productivity, supply chain, training, manpower optimization etc. These Kaizen improvement efforts are discussed in table 4 .

kaizen improvement.

Serial	Area of problem	Problem Description, and Kaizen improvement
1	Painting	Problem: Metal painting is poor adhesion of the paint to the surface. Kaizen Improvement: Introduce a practice of cleaning of metal with sandpaper and metal wire brush will make the paint adhere better. This will save rework cost Rs 600 (Refer annexure B)
2	Rust Formation	Problem: If the metal surface is not properly protected, rust can form underneath the paint, leading to unsightly stains and compromised durability of the paint job. Kaizen Improvement: Remove any existing rust by sanding or using a rust remover before painting. Apply a rust-inhibiting primer to prevent further rust formation. This will save rework cost Rs 600 (Refer annexure B)  Figure 6: Rusted part
3	Assembly Line	Problem: Supply Chain Disruptions Delays or shortages in the supply chain can halt production on

		the assembly line, leading to inefficiencies and missed deadlines. Kaizen Improvement: Maintain strong relationships with suppliers and implement inventory management systems to anticipate and mitigate potential disruptions. This will reduce the requirement of raw material inventory.
4	Equipment Malfunctions	Problem: Machinery breakdowns or malfunctions can cause production delays and impact overall productivity on the assembly line. Kaizen Improvement: Implement regular maintenance schedules for all equipment and machinery to prevent breakdowns. Data-driven Schedules Preventive Maintenance Training & Root Cause Analysis (annexure C)  Figure 7: Drill press



Figure 8: Cutting lathe Machine

5	Poorly Trained Personnel	Problem: Inadequately trained workers may struggle to perform their tasks efficiently, leading to errors, rework, and decreased productivity. Kaizen Improvement: Provide comprehensive training programs for assembly line workers to ensure they understand their roles, responsibilities, and procedures Instructor-Led Training On-the-Job Training
6	Man power optimization	Problem: Underutilization of Manpower: Some employees might not be fully engaged or utilized, leading to inefficiencies. Kaizen Improvement: Utilize workforce planning techniques to forecast staffing needs accurately, minimizing overstaffing or understaffing situations. This will save weekly Rs. 104000 (annexure C)

6. Draw the Future Value Stream Map

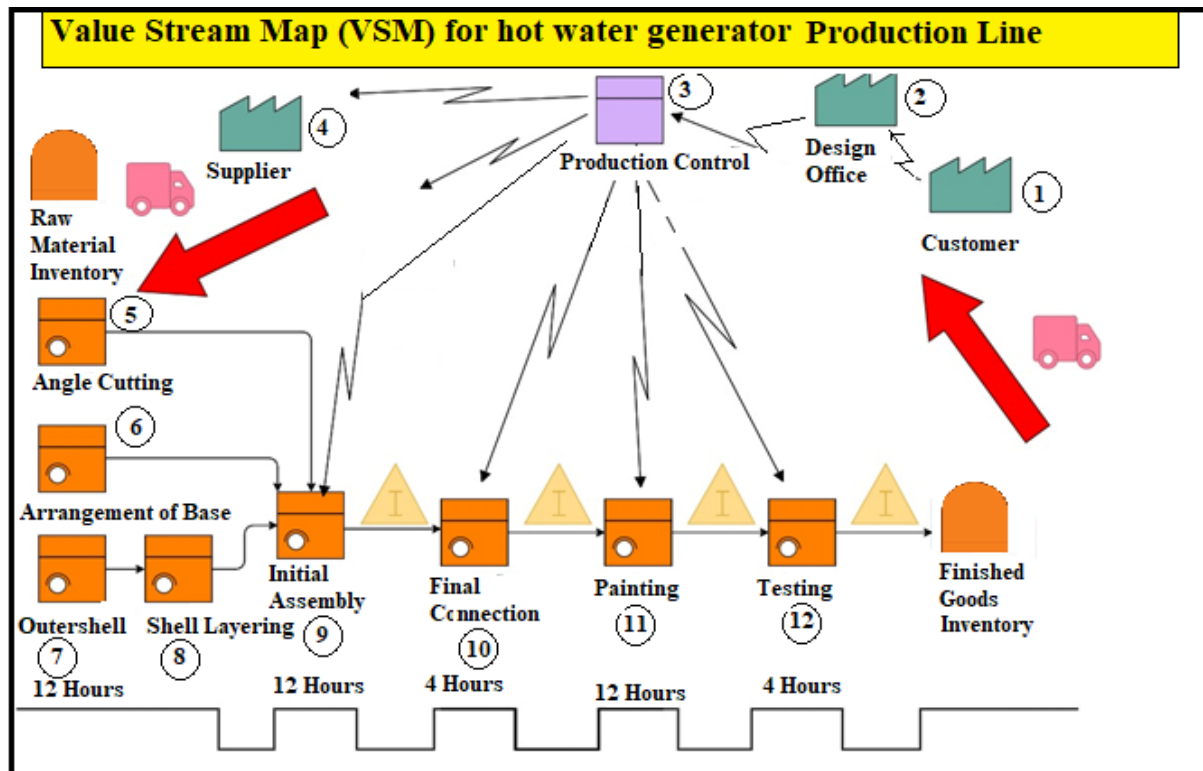


Figure 9: future Value Stream Map for the hot water generator production line.

Before	After
Number of workers=18	Number of workers=10
Number of failure/repair=13	Number of failure/repair=7

RESULT and DISCUSSION

The value stream mapping a modelling tool has been used to identify the gaps in the production line of hot water generator. However the system dynamics is added to the project in order to perform the dynamics snapshot. The integration of Value stream mapping and computer based simulation is used to get more compressive understanding as it provided closed loop systemic approach for complex production system. This paper provides practical framework called VSM-to-SD to assist modellers in efficiently translating VSM into quantitative SD models. The framework provided includes: 3.1 Develop Organizational Vision, 3.2 Draw the Current Value Stream Map, 3.3 Model the Current Value Stream using System Dynamics Framework, 3.4 Evaluate the Current Value Stream , 3.5 Prioritize Kaizen Improvement Ideas, 3.6 Draw the Future Value Stream Map.

The bottle necks were identified in the process :Painting and Rust formation, assembly line, equipment malfunction, poorly trained personnel.

- Kaizen improvement in each of the production gaps that were identified is provided :
- Painting- Introduce a practice of cleaning of metal (saves rework cost Rs 600)
- Rust formation- Remove any existing rust by sanding or using a rust remover before painting (saves rework cost Rs 600)
- Assembly line- in order to avoid Supply Chain Disruptions Delays or shortages, Maintain strong relationships with suppliers and implement inventory management systems
- Equipment Malfunction- Implemented regular maintenance schedules in order to avoid Machinery breakdowns or malfunctions that can cause production delays.
- Poorly Trained Personnel- Instructor-Led Training, On-the-Job Training comprehensive training programs are provided to Inadequately trained workers

- Man power optimization- The takt time graph and optimization of worker is used for forecast staffing needs accurately, minimizing overstaffing or understaffing situations (save weekly Rs. 104000).

CONCLUSION

The use of strategists plus decision-makers in Industry 4.0 plus Digital Change commonly count on decision-support systems to improve their logical abilities when analysing numerous choices for enhancing worth stream efficiency. Devices like Value Stream Mapping (VSM) as well as simulation have actually gotten acknowledgment for their useful applications as well as study payments in this area. VSM is a reputable technique in Operations Management coupled with Industrial Engineering due to the fact that it gives a simple and also global language for mapping procedures permitting experts together with scholars to interact successfully. On the various other hand, simulation (system dynamics - VENSIM), while a lot more intricate and also needing added modelling experience, supplies vibrant situations that can forecast prospective future efficiency end results.

This paper presents a structure VSM-SD. By utilizing a simple procedure and also usual modelling symbols, this structure assists in the fostering of SD for those currently knowledgeable about VSM. The outcomes were created promptly permitting supervisors to make educated choices concerning production line, therefore minimizing the gaps connected with decision-making.

In addition, this strategy can be related to different production contexts and also solution markets like Lean Healthcare where Lean Thinking is being taken on. An in-depth overview might likewise be created to assist VSM specialists with restricted modelling experience incorporate SD right into their worth stream evaluation motivating wider fostering of the structure.

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Annexure A

Table 5: Explanation of lean box score

Lean Box Score Parameter	Value	Explanation
Dock to dock days:	58 Hours	This is obtained from the value stream map by totaling the lead time , and cycle time
First Time Through (FTT)	75%	During the Gemba walk it was noted that 25% of products do go through re-work.
On-Time Delivery (OTD)	100%	All the products are delivered on time.
Sales per person	Rs 130,000	$(\text{Number of items produced per week}) \times (\text{Price of one item}) / (\text{Number workers employed in the value stream}) = ((208/52) \times (585000))/18$
Average cost per unit	Rs 296982	$\text{Average cost per unit} = (\text{Conversion cost per week} + \text{Inventory Cost per week} + \text{Material cost per week}) / \text{number of pieces produced in week} = (87839 + 88 + 1100000)/4 = 296982$
Resource Capacity (productive)	Paint Guns - $5/36=0.13$ Drilling nuts- $24/36=0.66$ Lifting machine- $12/36=0.33$ Welding machine- $30/36=0.83$ Shearing machine- $20/36=0.55$ Bending machine- $15/36=0.41$	Productive time = Machine working time/ Available time
Resource Capacity (non-productive)	Paint Guns- $0.166/36=0.004$ Drilling nuts- $0.083/35=0.002$ Lifting machine- $0.5/36=0.016$ Welding machine- $0.166/36=0.004$ Shearing machine- $0.5/36=0.016$	Machine setup or cleaning time/ Available time

	Bending machine- $0.5/36=0.016$	
Resource Capacity Available	Paint Guns- $1 - 0.13 - 0.004 = 0.866$ Drilling nuts- $1 - 0.66 - 0.002=0.338$ Lifting machine- $1 - 0.33 - 0.016=0.654$ Welding machine- $1 - 0.83 - 0.004=0.166$ Shearing machine- $1 - 0.55 - 0.016=0.434$ Bending machine- $1 - 0.41 - 0.016=0.574$	Resource Capacity Available = (1 – productive – non-productive)

Conversion cost is the costs incurred in converting the raw materials into final products. This cost is the sum of the following components: (1) Ownership Cost of machine (2) Running cost of the machine (3) Cost of manpower employed in running the machine.

All these costs have been discussed in table X.

Ownership of the cost of the machine = Cost of machine per week = Initial cost / 5 years x 52. This assumes the life of machine is 5 years (refer table y).

Manpower cost is weekly wages of the workers employed on the machine (refer table y).

Running cost includes electricity cost, and cost of any consumable used on the machine (table z).

Table 6: Explanation of Conversion cost

Mach ine name	Ownership Cost of machine (1) (rupees)(fixed)(annual) (A)	Ownership Cost of machine (1) (rupees)(fixed)(weekly) (B) = (A)/52	Running cost of machine (variable)(weekl y)(*)(C)	Manpowe r Cost (Number of workers x weekly cost per worker(37 50 rupee))	Conversion cost=runnin g cost +cost of machine per week+manp ower cost (D)
Paint gun	1,00,000	$100000/52=1923$	1521.3	$3*3750=11250$	14694.3
Drilli ng nut	40,000	$40000/52=769.2$	59.64	$1*3750=3750$	4578.84
Liftin g machi ne	6,00,000	$600000/52=11538.4$	9087.3	$2*3750=7500$	28125.7

Welding machine	35,000	$35000/52=673$	11200	$2*3750=7500$	19373
Shearing machine	1,50,000	$150000/52=2884.6$	170	$2*3750=7500$	10554.6
Bending machine	1,50,000	$150000/52=2884.6$	127.8	$2*3750=7500$	10512.4
Total		20672.8	22166.04	45000	87838.84

Explanations for the running cost (C) are given in table X given in table z.

Table 7: Running cost

Machine name	Explanation for Running Cost (variable)
Paint gun	Electricity Cost = Wattage of machine x Number of hours used per week x Cost of per unit electricity = 0.6 Kw x 5 Hour x Rs 7.1 per unit = Rs 21.3 Cost of Paint = Price per lit x number of liters consumed per week = 10 x 150 = Rs 1500
Drilling nut	Electricity Cost = Wattage of machine x Number of hours used per week x Cost of per unit electricity = 0.35Kw x 24 Hour x Rs 7.1 per unit = Rs 59.64
Lifting machine	Fuel Cost = Fuel (Liters) used per hour x Number of hours used x cost per litres= 8 Liters per hour x 12 Hour per week x Rs 94.66 per liter = Rs 9087.3
Welding machine	Electricity Cost = Wattage of machine x Number of hours used per week x Cost of per unit electricity = 1Kw x 30 Hour x Rs 7.1 per unit = Rs 213 Gas Cost = Volume of Gas x Cost of Gas = 10 x 28 x 40 = Rs 11200
Shearing machine	Electricity Cost = Wattage of machine x Number of hours used per week x Cost of per unit electricity = 1.2 Kw x 20 Hour x Rs 7.1 per unit = Rs 170
Bending machine	Electricity Cost = Wattage of machine x Number of hours used per week x Cost of per unit electricity = 1.2 Kw x 15 Hour x Rs 7.1 per unit = Rs 127.8

Conversion cost=63051.84

Value stream profit

$$\text{Value stream profit} = \text{Revenue} - \text{Average cost per unit}$$

Revenue=2340000, Average cost per unit=296982

$$2340000-296982=2043018$$

Annexure B

This annexure discusses problem of poor quality of surface, which leads to paint peeling off. Cleaning it thoroughly to remove dirt, grease, rust, and other contaminants. Use sandpaper or a wire brush to roughen the surface slightly, which will help the paint adhere better.

Due to poor quality of surface having dirt, moisture etc. paint peels off. This result in an average rework of two hours per week.

Machine cost per hour= Rs $100000/(52*7*24)$ = Rs 12

Cost of rework = Hours of rework x (manpower cost per hour + Machine cost per hour) = 2 x (96 x 3 + 12) =600

Annexure - C

Manpower optimization is a broad strategy used across various industries to ensure you have the right number of people with the right skills in the right place at the right time.

The takt time represents the average rate at which units need to be produced to meet customer demand. It's calculated by dividing the Total Available Production Time by the Customer Demand Rate.

Takt time line = available hours/ number of pieces per week = $36/4 = 9$

Broadly speaking, if the cycle time is less than takt time, then we have a room to deploy resources from that production station, to the production station where cycle time is more than takt time. In the figure 10, activities 1 to 8 have been shown. The height of the bar represents cycle time. The grey horizontal line represents the takt time. The number in the square bracket represents the number of workers employed at the production station. For example, “initial assembly” production station employs three workers. From the outset, we can guess that workers from the station 1, 3,4,6,8 can be re-deployed at station 2, 5,7. However, it requires a careful analysis. Initially there are 18 workers, named by letters: A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R. These workers have been re-deployed. The analysis is shown in table 8.

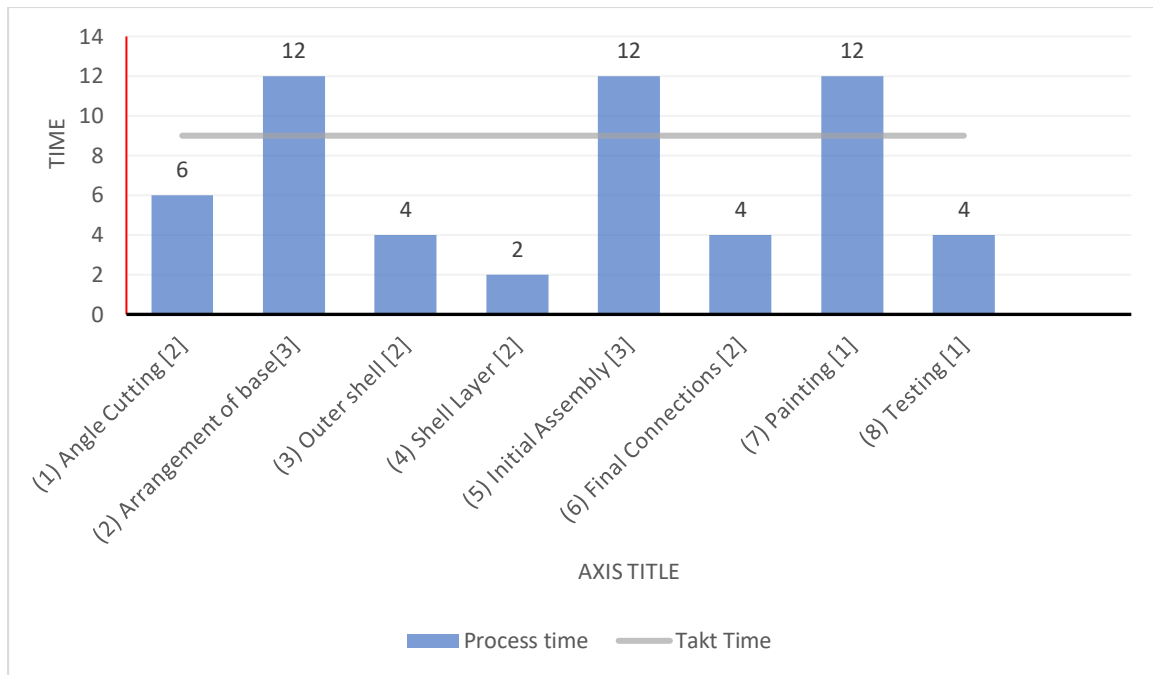


Figure 10: manpower optimization using takt time approach

Table 8: Optimization of workers

Proce ss name	Angl e Cutti ng	Arrangem ent of base	Out er shel l	She ll Lay er	Initial Assem bly	Final Connecti ons	Painti ng	Testi ng
Cycle time(h r)	6	12	4	2	12	4	12	4
No. of worker s	2	3	2	2	3	2	3	1
Namin g of worker (assu med name)	A,B	C,D,E	F,G	H,I,	J,K,L	M,N	O,P, Q	R
Optimi zed worker allotm ent	A,B	C,D,E	A,B	A,B	C,D,E	M,N	O,P, Q	A

After optimization 10 workers are required: A,B,C,D,E,M,N,O,P,Q .

Following strategies are used:

- Cross-Training: Train workers on multiple tasks so they can be flexibly deployed during peak and low demand periods.
- Temporary Staffing: Consider hiring temporary workers during peak demand periods to avoid overworking existing staff.
- Work-Sharing: During slow periods, consider work-sharing programs to distribute available work among current employees and avoid layoffs.

Sales per person	
Before	After
Number of workers=18	Number of workers=10
$(208 * 585000)/(52 * 18) = 130,000$	$(208 * 585000)/(52 * 10) = 234000$